

DISTURBANCES

*all active during training
(domain randomization)*

Weather (η)

heavy-tailed delays

Breakdowns (λ)

a bus is removed

Demand surge (σ)

crowding at stops

Traffic (σ -)

variable cruise speed

*New seed each episode, so
the policy meets diverse
conditions*

DECISION & LEARNING LOOP

repeated at every control event

Bus arrives at a control stop

1 • Observe state s_i

headways h^- , $\hat{h}^+$ • load • waiting • disturbance flags

2 • Shared policy $\pi\theta$ decides

one network used by all buses (parameter sharing)

3 • Act a_i

holding (α) and/or stop-skipping — 10 discrete actions

4 • Receive reward r_i

rewards even headways & short waits; discounted $e^{-\beta t}$

5 • Update the policy (DDQN)

store (s, a, r, s') in replay buffer $\rightarrow$ improve θ

next agent / next event

OUTPUT OF TRAINING

A trained policy θ

experienced weather, breakdowns,
surges & traffic repeatedly

Frozen for evaluation

weights fixed; handed to
the evaluation layer

*No performance metrics are
logged in this phase, the
loop only updates θ .*